

The Unified Three-Republic Theorem

Algebraic Consistency, Strategic Welfare, Dynamic Stability,
Latent-State Identification, Preventive Diagnostics, and Robust Control

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Abstract

This paper develops a unified mathematical framework for a system of three interacting Republics in which algebraic consistency, strategic optimality, welfare efficiency, dynamic stability, latent-state identification, preventive diagnostics, parameter uncertainty, and stochastic shocks are treated as components of one equilibrium object. The central state is $s = (r_1, r_2, r_3, p)^\top$, where r_i denotes the structural state of Republic i and p denotes a common political-institutional state. The three Republic states admit a permutation-invariant representation through the elementary symmetric polynomials r_A , r_B , and r_C , which define a cubic whose roots are the Republic-specific states. Strategic interaction is represented by a best-response map R , and the equilibrium dynamics are represented by a reduced transition map T^* . We derive the strategic feedback multiplier $(I - D_u R)^{-1}$ and show that the complete equilibrium Jacobian contains both direct dynamic feedback and strategic amplification. This permits a distinction between strategic fragility, common-mode dynamic fragility, and asymmetric-mode fragility. A welfare layer is introduced through weighted Republic objectives and constrained second-order conditions. Latent states are required to be identifiable from the observable process, with identification formulated through observability rank and singular-value margins. Regime I is divided into robust, warning, and endangered subregions, so that instability can be diagnosed before an actual eigenvalue crosses the unit circle. The framework is then extended to stochastic shocks and parameter uncertainty through chance constraints, robust stability margins, and a robust viability kernel. Finally, a minimum-cost preventive intervention problem is formulated, producing a closed-loop system that observes, identifies, diagnoses, forecasts, intervenes, and re-identifies. Connections to economics, finance, physics, chemistry, biology, psychology, psychiatry, political science, geopolitics, diplomacy, international relations, warfare economics, statistics, computer science, algorithms, data science, and artificial intelligence are discussed as applications and interpretations of the common mathematical architecture rather than as substitutes for mathematical proof.

The paper ends with “The End”

Keywords: three-Republic system; equilibrium theory; dynamic games; welfare economics; strategic feedback; systemic risk; latent states; observability; robust control; viability theory; early warning systems; algebraic invariants; stochastic stability; political economy; geopolitics; computational economics.

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1 Introduction

Many models of political economy, international relations, finance, and strategic interaction separate questions that are in reality simultaneous. A model may ask whether an algebraic equilibrium exists, while another asks whether agents have optimal strategies, another asks whether the resulting trajectory is dynamically stable, and another asks whether the latent state driving the observed data can actually be identified. A fifth model may address systemic risk only after instability has occurred.

The purpose of the present framework is to impose a stronger standard. A valid equilibrium of a three-Republic system should satisfy all of the following:

1. algebraic consistency;
2. strategic optimality;

3. welfare-strategic compatibility;
4. dynamic stability;
5. latent-state identification;
6. institutional admissibility;
7. robustness to parameter uncertainty;
8. resilience to stochastic shocks;
9. and, where possible, existence of a preventive control policy.

The resulting object is not merely a Nash equilibrium, not merely a fixed point, and not merely a statistical latent-state estimate. It is a constrained equilibrium that is mathematically coherent, strategically admissible, dynamically stable, welfare-consistent, empirically recoverable, and preventively controllable.

The central state is

$$s_t = \begin{pmatrix} r_{1t} \\ r_{2t} \\ r_{3t} \\ p_t \end{pmatrix}. \quad (1)$$

The system contains controls

$$u_t = \begin{pmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{pmatrix}, \quad (2)$$

and potentially unobserved structural states z_t .

The complete stochastic system is

$$s_{t+1} = T(s_t, u_t, \theta, \varepsilon_{t+1}), \quad (3)$$

$$u_t^* = R(u_t^*, s_t, a_t, \theta, \xi_{t+1}), \quad (4)$$

$$z_{t+1} = F_z(z_t, s_t, u_t, \theta, \eta_{t+1}), \quad (5)$$

and

$$Y_t = H(s_t, z_t, \theta) + \nu_t. \quad (6)$$

Here θ denotes structural parameters, a_t denotes preventive intervention, and the disturbances ε_t , ξ_t , η_t , and ν_t represent distinct sources of uncertainty.

The central thesis is that these equations should not be treated as independent modules. They form a coupled dynamical game.

2 Conceptual Architecture

The framework can be represented schematically as follows.

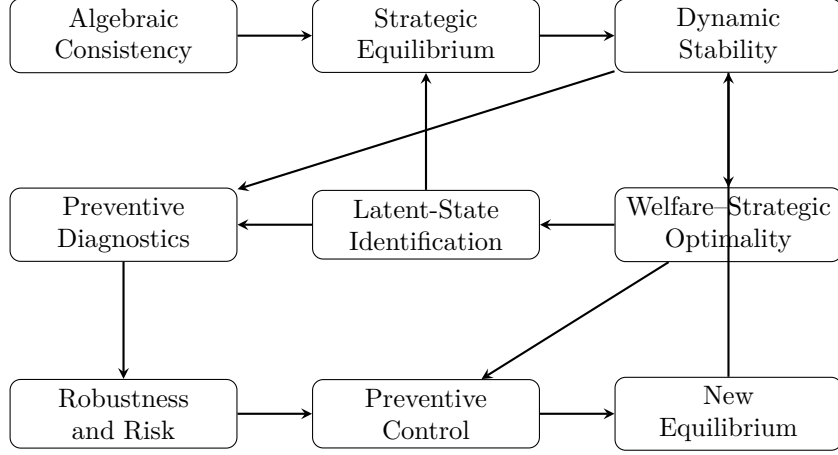


Figure 1: Conceptual architecture.

The system therefore implements the closed loop

$$\boxed{\text{observe} \rightarrow \text{identify} \rightarrow \text{diagnose} \rightarrow \text{forecast} \rightarrow \text{intervene} \rightarrow \text{re-identify} \rightarrow \text{re-diagnose.}} \quad (7)$$

This closed loop is the principal extension beyond a static equilibrium theorem.

3 Algebraic Structure of the Three-Republic System

3.1 Republic-specific states

Let

$$r = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}. \quad (1)$$

Define the elementary symmetric invariants

$$r_A = r_1 + r_2 + r_3, \quad (8)$$

$$r_B = r_1 r_2 + r_1 r_3 + r_2 r_3, \quad (9)$$

and

$$r_C = r_1 r_2 r_3. \quad (10)$$

The three Republic states are therefore roots of

$$P(\lambda) = \lambda^3 - r_A \lambda^2 + r_B \lambda - r_C. \quad (11)$$

Thus

$$P(r_i) = 0, \quad i = 1, 2, 3. \quad (12)$$

This representation is invariant under permutations of the Republic labels.

Lemma 3.1 (Algebraic recoverability). *Suppose (r_A, r_B, r_C) is known and the polynomial P in equation (11) has three admissible real roots. Then the unordered set*

$$\{r_1, r_2, r_3\} \quad (2)$$

is uniquely determined.

Proof. The coefficients of a monic polynomial uniquely determine its multiset of roots over the complex numbers, including multiplicity. Since the coefficients in equation (11) are the elementary symmetric functions of r_1, r_2, r_3 , the root multiset is uniquely determined. Label-specific recovery requires an additional labeling condition. $\square$

3.2 Discriminant

The cubic discriminant is

$$\Delta = r_A^2 r_B^2 - 4r_B^3 - 4r_A^3 r_C - 27r_C^2 + 18r_A r_B r_C. \quad (13)$$

For three distinct real roots,

$$\Delta > 0. \quad (14)$$

When

$$\Delta = 0, \quad (15)$$

at least two Republic states coincide.

A normalized algebraic separation margin can be defined by

$$m_\Delta = \frac{\Delta}{(1 + |r_A| + |r_B| + |r_C|)^6}. \quad (16)$$

The precise normalization is not unique; its purpose is to reduce scale dependence.

Remark 3.2. *The discriminant should be interpreted as an algebraic state-separation statistic, not as a universal measure of political or economic divergence. Its empirical meaning depends on the interpretation assigned to r_i .*

4 Algebraic Equilibrium Restrictions

Let the structural restrictions be

$$\Phi(s, u; \theta) = 0. \quad (17)$$

The admissible equilibrium set is

$$\mathcal{A} = \{(s, u) : \Phi(s, u; \theta) = 0, C_j(s, u; \theta) \geq 0, j = 1, \dots, m\}. \quad (18)$$

Here the functions C_j may represent resource, institutional, feasibility, probability, budgetary, legal, or strategic constraints.

Suppose Republic-specific structural maps are given by f , g , and h . The previously considered coincidence condition is

$$f(r_f, p) = g(r_f, p) = h(r_f, p) \iff (r_f, p) = (0, 0). \quad (19)$$

Equation (19) should not be confused with the complete equilibrium condition. It characterizes a pointwise coincidence of the Republic-specific maps. The general equilibrium instead satisfies

$$r_i^* = R_i(r_{-i}^*, p^*), \quad p^* = \Phi_p(r^*, p^*, u^*), \quad (20)$$

together with the remaining constraints.

5 Strategic Equilibrium

Let Republic i have objective

$$V_i(u_i, u_{-i}, s) = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_i^t [W_i(s_t) - C_i(u_{it}, u_{-it}, s_t)]. \quad (21)$$

Define the best-response correspondence

$$BR_i(u_{-i}, s) = \arg \max_{u_i \in \mathcal{U}_i} V_i(u_i, u_{-i}, s). \quad (22)$$

A strategic equilibrium satisfies

$$u_i^* \in BR_i(u_{-i}^*, s^*), \quad i = 1, 2, 3. \quad (23)$$

When best responses are single-valued, write

$$u^* = R(u^*, s). \quad (24)$$

Assumption 5.1. *For each i , $\mathcal{U}_i$ is nonempty, compact, and convex, while V_i is continuous in all arguments and quasi-concave in u_i .*

Proposition 5.2 (Strategic existence). *Under the preceding assumption, a strategic equilibrium exists for every fixed admissible state s .*

Proof. Continuity and compactness imply that each best-response correspondence is nonempty and compact-valued. Quasi-concavity gives convex-valuedness. Upper hemicontinuity follows from standard maximum-theorem arguments. The product correspondence therefore satisfies the hypotheses of Kakutani's fixed-point theorem, yielding a fixed point. $\square$

6 Strategic Uniqueness and Amplification

Let

$$K = D_u R. \quad (25)$$

For the three-Republic case,

$$K = \begin{pmatrix} 0 & a & b \\ c & 0 & d \\ e & f & 0 \end{pmatrix}. \quad (26)$$

A sufficient contraction condition is

$$\|K\|_\infty = \max_i \sum_{j \neq i} |K_{ij}| < 1. \quad (27)$$

Proposition 6.1 (Strategic uniqueness). *If*

$$\|D_u R\|_\infty \leq \kappa_S < 1, \quad (28)$$

then $R(\cdot, s)$ is a contraction and the strategic equilibrium is unique for every admissible s .

Proof. The contraction mapping theorem gives existence and uniqueness. Iteration of the best-response map converges geometrically at rate at most κ_S . $\square$

The exact determinant of the strategic feedback system is

$$\det(I - K) = 1 - ac - be - df - ade - bcf. \quad (29)$$

The terms ac , be , and df represent pairwise feedback cycles. The terms ade and bcf represent directed three-Republic cycles.

Hence the three-Republic system contains feedback structures not present in a purely bilateral model. Whenever

$$\rho(K) < 1, \quad (30)$$

we have the Neumann expansion

$$(I - K)^{-1} = I + K + K^2 + K^3 + \dots. \quad (31)$$

The matrix $(I - K)^{-1}$ is therefore a strategic feedback multiplier.

7 The Strategic Feedback Identity

Let the primitive transition map be

$$\tilde{T}(s, u; \theta). \quad (32)$$

After strategic optimization,

$$T^*(s; \theta) = \tilde{T}(s, u^*(s; \theta); \theta). \quad (33)$$

Differentiating

$$u^* = R(u^*, s) \quad (3)$$

gives

$$D_s u^* = D_u R D_s u^* + D_s R. \quad (34)$$

Hence

$$\boxed{D_s u^* = (I - D_u R)^{-1} D_s R.} \quad (35)$$

Substitution into equation (33) yields

$$\boxed{D_s T^* = D_s \tilde{T} + D_u \tilde{T} (I - D_u R)^{-1} D_s R.} \quad (36)$$

Theorem 7.1 (Strategic amplification theorem). *Suppose $I - D_u R$ is nonsingular. Then the reduced equilibrium Jacobian is given by equation (36). Consequently, dynamic stability depends not only on direct state feedback but also on strategic amplification through $(I - D_u R)^{-1}$.*

Proof. Equation (35) follows by rearranging equation (34). Applying the chain rule to equation (33) gives

$$D_s T^* = D_s \tilde{T} + D_u \tilde{T} D_s u^*. \quad (4)$$

Substituting equation (35) yields equation (36). $\square$

A sufficient stability condition is

$$\boxed{\|D_s \tilde{T}\| + \|D_u \tilde{T}\| \frac{\|D_s R\|}{1 - \|D_u R\|} < 1.} \quad (37)$$

Equation (37) demonstrates how a system can become fragile because strategic interactions amplify otherwise moderate direct dynamics.

8 Dynamic Stability

Let

$$s_{t+1} = T^*(s_t). \quad (38)$$

A stationary equilibrium satisfies

$$s^* = T^*(s^*). \quad (39)$$

The local Jacobian is

$$J^* = D_s T^*(s^*). \quad (40)$$

For discrete-time dynamics, local asymptotic stability requires

$$\rho(J^*) < 1. \quad (41)$$

Writing

$$s_t = s^* + \delta_t, \quad (5)$$

we obtain

$$\delta_{t+1} = J^* \delta_t + O(\|\delta_t\|^2). \quad (42)$$

Thus sufficiently small perturbations decay when equation (41) holds.

8.1 Block decomposition

Write

$$s = \begin{pmatrix} r \\ p \end{pmatrix}, \quad (6)$$

and

$$J^* = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (43)$$

Then

$$\det(\lambda I - J^*) = \det(\lambda I - D) \det[\lambda I - A - B(\lambda I - D)^{-1}C]. \quad (44)$$

The effective Republic-state Jacobian is therefore

$$A_{\text{eff}}(\lambda) = A + B(\lambda I - D)^{-1}C. \quad (45)$$

This establishes the role of common political-institutional feedback in altering the effective stability of the Republic subsystem.

9 Common and Asymmetric Modes

In a symmetric three-Republic system, define

$$r_t = \bar{r}_t \mathbf{1} + r_t^\perp, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (7)$$

with

$$\mathbf{1}^\top r_t^\perp = 0. \quad (46)$$

The common mode is proportional to $\mathbf{1}$. The asymmetric subspace is the two-dimensional orthogonal complement.

For

$$K = \begin{pmatrix} 0 & \alpha & \beta \\ \beta & 0 & \alpha \\ \alpha & \beta & 0 \end{pmatrix}, \quad (47)$$

the eigenvalues are

$$\mu_0 = \alpha + \beta, \quad (48)$$

and

$$\mu_\pm = -\frac{\alpha + \beta}{2} \pm i\frac{\sqrt{3}}{2}(\alpha - \beta). \quad (49)$$

Moreover,

$$|\mu_\pm|^2 = \alpha^2 - \alpha\beta + \beta^2. \quad (50)$$

Thus

$$\rho(K) = \max \left\{ |\alpha + \beta|, \sqrt{\alpha^2 - \alpha\beta + \beta^2} \right\}. \quad (51)$$

The common and asymmetric modes can therefore be diagnosed separately. Define

$$m_C = 1 - \rho(J_C), \quad (52)$$

and

$$m_A = 1 - \rho(J_A). \quad (53)$$

Then

$$m_C > 0, \quad m_A > 0 \quad (54)$$

defines dynamically stable Regime I.

10 Four Dynamic Regimes

The basic regime classification is

Regime	Common mode	Asymmetric mode
I	Stable	Stable
II	Unstable	Stable
III	Stable	Unstable
IV	Unstable	Unstable

Table 1: Basic regime classification.

Regime II represents common systemic instability. Regime III represents Republic divergence or asymmetric instability. Regime IV represents compound instability.

This decomposition is more informative than a single aggregate stability statistic because instability can occur in the asymmetric subspace while aggregate variables remain apparently well behaved.

11 Welfare and Strategic Optimality

Let welfare weights satisfy

$$\omega_i > 0, \quad i = 1, 2, 3. \quad (55)$$

Define

$$W^\omega = \sum_{i=1}^3 \omega_i V_i. \quad (56)$$

At an interior optimum,

$$\nabla_u W^\omega = 0. \quad (57)$$

For strict local optimality,

$$D_{uu}^2 W^\omega \prec 0. \quad (58)$$

Under constraints, the appropriate object is the Hessian restricted to the tangent space of the active constraints.

Let T_C denote that tangent space. A sufficient second-order condition is

$$v^\top D^2 W^\omega v < 0 \quad \forall v \in T_C \setminus \{0\}. \quad (59)$$

11.1 Welfare externalities

Define

$$X_{ij} = \frac{\partial W_i}{\partial r_j}. \quad (60)$$

A weighted welfare balance satisfies

$$\omega^\top X = 0. \quad (61)$$

This condition captures the possibility that a Republic's welfare depends on the states of the other Republics.

11.2 Strategic welfare curvature

Because

$$u^* = u^*(s), \quad (8)$$

the reduced welfare function is

$$\mathcal{V}(s) = W^\omega(s, u^*(s)). \quad (62)$$

Its gradient is

$$D_s \mathcal{V} = D_s W^\omega + D_u W^\omega D_s u^*. \quad (63)$$

At an interior strategic welfare optimum,

$$D_u W^\omega = 0, \quad (9)$$

so

$$D_s \mathcal{V} = D_s W^\omega. \quad (64)$$

The second derivative contains the strategic-response term

$$(D_s u^*)^\top D_{uu}^2 W^\omega D_s u^*. \quad (65)$$

Hence strategic interaction affects welfare curvature even when first-order strategic terms vanish.

12 Pareto Efficiency on the Unified Manifold

A feasible perturbation is Pareto improving if

$$\Delta V_i > 0 \quad \forall i. \quad (66)$$

The relevant feasible set, however, is not the unrestricted policy space. It is the unified equilibrium manifold defined by algebraic, strategic, dynamic, and institutional restrictions.

Proposition 12.1 (Local Pareto efficiency). *Suppose the weighted welfare objective has positive weights, satisfies the first-order equilibrium conditions, and is strictly concave on the tangent space of the unified constraints. Then the equilibrium is locally Pareto efficient within the unified equilibrium manifold.*

Proof. If a feasible Pareto improvement existed, the weighted welfare objective would strictly increase because every $\omega_i > 0$. This contradicts the local maximum implied by the second-order condition. $\square$

13 Latent-State Identification

Let

$$z_t \in \mathbb{R}^k \quad (10)$$

be a latent structural state.
Observations satisfy

$$Y_t = H(q_t, z_t, \theta) + \nu_t, \quad (67)$$

where

$$q_t = \begin{pmatrix} r_{A,t} \\ r_{B,t} \\ r_{C,t} \\ p_t \end{pmatrix}. \quad (68)$$

Two latent states are observationally equivalent if they induce the same observable probability law.
Write

$$z \sim_{\text{obs}} \tilde{z}. \quad (69)$$

Identification means

$$z \sim_{\text{obs}} \tilde{z} \implies z = \tilde{z}, \quad (70)$$

or, when a symmetry group Γ is intrinsic to the model,

$$z \sim_{\text{obs}} \tilde{z} \implies \tilde{z} = \gamma(z) \quad \text{for some } \gamma \in \Gamma. \quad (71)$$

The three-Republic system naturally contains a permutation symmetry unless institutional labels are directly observed.

13.1 Local observability

For deterministic local dynamics

$$z_{t+1} = F_z(q_t, z_t), \quad (72)$$

construct

$$\mathcal{O}_z = \begin{pmatrix} H_z \\ H_z F_z \\ H_z F_z^2 \\ \vdots \\ H_z F_z^{k-1} \end{pmatrix}. \quad (73)$$

Proposition 13.1 (Local latent-state identification). *If*

$$\text{rank } \mathcal{O}_z(z^*) = k, \quad (74)$$

then the latent state is locally observable from sufficiently long observable histories.

The practical identification margin is

$$m_I = \sigma_{\min}(\mathcal{O}_z), \quad (75)$$

where $\sigma_{\min}$ denotes the smallest singular value.
Exact local observability requires

$$m_I > 0. \quad (76)$$

Practical identification becomes fragile as

$$m_I \rightarrow 0. \quad (77)$$

14 The Unified Equilibrium

Define the equilibrium residual

$$\mathcal{F}(s, u; \theta) = \begin{pmatrix} u - R(u, s; \theta) \\ \Phi(s, u; \theta) \\ T(s, u; \theta) - s \end{pmatrix}. \quad (78)$$

A regular equilibrium satisfies

$$\mathcal{F}(s^*, u^*; \theta) = 0. \quad (79)$$

The master Jacobian is

$$D_x \mathcal{F} = \begin{pmatrix} I - D_u R & -D_s R \\ \Phi_s & \Phi_u \\ T_s & T_u \end{pmatrix}, \quad (80)$$

where

$$x = \begin{pmatrix} s \\ u \end{pmatrix}. \quad (11)$$

A regular local equilibrium satisfies

$$\det D_x \mathcal{F} \neq 0 \quad (81)$$

when the system is square.

This is an equilibrium singularity condition. As the determinant approaches zero, local uniqueness and comparative-static stability may deteriorate.

15 Comparative Statics

Differentiating

$$\mathcal{F}(x^*; \theta) = 0 \quad (12)$$

with respect to θ_j gives

$$D_x \mathcal{F} \frac{\partial x^*}{\partial \theta_j} + \frac{\partial \mathcal{F}}{\partial \theta_j} = 0. \quad (82)$$

Hence

$$\boxed{\frac{\partial x^*}{\partial \theta_j} = -(D_x \mathcal{F})^{-1} \frac{\partial \mathcal{F}}{\partial \theta_j}}. \quad (83)$$

Consequently,

$$\|(D_x \mathcal{F})^{-1}\| \quad (84)$$

is a natural equilibrium-fragility measure.

A large value indicates that small parameter changes can generate large equilibrium responses.

16 Preventive Diagnostics

Ordinary stability classification is insufficient for prevention. The system should measure its distance from each failure boundary while it remains in Regime I.

Define the strategic margin

$$m_S = 1 - \rho(D_u R), \quad (85)$$

the common-mode margin

$$m_C = 1 - \rho(J_C), \quad (86)$$

the asymmetric-mode margin

$$m_A = 1 - \rho(J_A), \quad (87)$$

the identification margin

$$m_I = \sigma_{\min}(\mathcal{O}_z), \quad (88)$$

the algebraic margin m_Δ from equation (16), and the welfare curvature margin

$$m_W = \lambda_{\min}(-D^2 W^\omega|_{T_C}). \quad (89)$$

The diagnostic vector is

$$\mathcal{D}_t = \begin{pmatrix} m_S(t) \\ m_C(t) \\ m_A(t) \\ m_I(t) \\ m_\Delta(t) \\ m_W(t) \end{pmatrix}. \quad (90)$$

16.1 Diagnostic levels

For each margin choose thresholds

$$0 < \tau_j^{\text{red}} < \tau_j^{\text{amber}}. \quad (91)$$

Then define:

Level	Condition	Interpretation
Green	$m_j > \tau_j^{\text{amber}}$	Comfortable margin
Amber	$\tau_j^{\text{red}} < m_j \leq \tau_j^{\text{amber}}$	Preventive warning
Red	$0 < m_j \leq \tau_j^{\text{red}}$	Endangered but stable
Breach	$m_j \leq 0$	Boundary crossed

Table 2: Diagnostic levels.

The thresholds are policy parameters, not mathematical constants.

16.2 Deterioration rates

For a sampling interval Δt ,

$$\dot{m}_j(t) \approx \frac{m_j(t) - m_j(t - \Delta t)}{\Delta t}. \quad (92)$$

Define

$$D_j(t) = -\dot{m}_j(t). \quad (93)$$

Large positive D_j indicates rapid deterioration.

A simple local estimate of boundary-crossing time is

$$\tau_j^{\text{breach}} \approx \frac{m_j}{-\dot{m}_j}, \quad \dot{m}_j < 0. \quad (94)$$

The earliest predicted crossing is

$$\tau_{\min} = \min_j \tau_j^{\text{breach}}. \quad (95)$$

17 Regime I as a Preventive Region

The ordinary stable region is

$$\mathcal{S}_I = \{s : m_C > 0, m_A > 0\}. \quad (96)$$

Define the robust safe region

$$\mathcal{S}_{\text{safe}} = \{s : m_j(s) \geq \bar{m}_j, \quad j \in \{S, C, A, I, \Delta, W\}\}. \quad (97)$$

Then

$$\mathcal{S}_{\text{safe}} \subseteq \mathcal{S}_I. \quad (98)$$

Regime I is therefore subdivided into

$$I_0 = \text{robust Regime I}, \quad (99)$$

$$I_1 = \text{strategic warning}, \quad (100)$$

$$I_2 = \text{common-mode warning}, \quad (101)$$

$$I_3 = \text{asymmetric-mode warning}, \quad (102)$$

and

$$I_4 = \text{compound warning}. \quad (103)$$

These states remain mathematically inside Regime I provided

$$m_C > 0, \quad m_A > 0. \quad (104)$$

The transition graph is

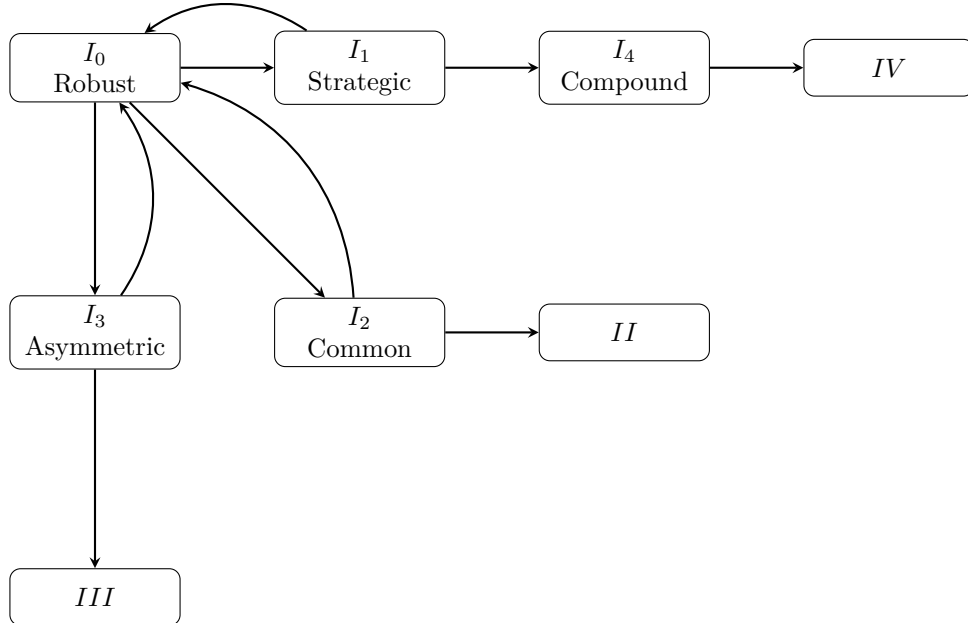


Figure 2: Transition graph.

Preventive policy seeks to reverse warning states toward I_0 before the system reaches II , III , or IV .

18 Transient Stability and Non-Normality

Spectral stability is not equivalent to absence of transient amplification.

Even if

$$\rho(J) < 1, \quad (105)$$

a non-normal matrix may satisfy

$$\|J^t\| \gg 1 \quad (13)$$

for intermediate t .

Define

$$\mathcal{A}_T = \max_{1 \leq t \leq T} \|J^t\|. \quad (106)$$

This provides a transient-resilience diagnostic.

The distinction is

$$\boxed{\text{asymptotic stability} \neq \text{short-run resilience.}} \quad (14)$$

This distinction has direct analogues in finance, where a system may eventually mean-revert while experiencing a severe temporary drawdown; in physics, where a linearly stable system can exhibit transient energy amplification; and in ecology, where a stable equilibrium can nevertheless suffer a large transient population collapse.

19 Stochastic Shocks and Parameter Uncertainty

Let

$$\theta \in \Theta \quad (15)$$

be uncertain, and let shocks belong to an admissible distribution class $\mathcal{P}$.

The stochastic transition is

$$s_{t+1} = T(s_t, u_t, \theta, \varepsilon_{t+1}). \quad (107)$$

A one-period chance constraint is

$$\Pr [m_j(s_{t+1}) \geq \bar{m}_j \mid s_t, a_t] \geq 1 - \alpha_j. \quad (108)$$

If

$$\sum_j \alpha_j \leq \alpha, \quad (109)$$

the union bound gives

$$\Pr [m_j(s_{t+1}) \geq \bar{m}_j \ \forall j] \geq 1 - \alpha. \quad (110)$$

19.1 Robust margins

Define

$$\underline{m}_j(s) = \inf_{\theta \in \Theta} m_j(s; \theta). \quad (111)$$

Robust dynamic stability requires

$$\sup_{\theta \in \Theta} \rho(J(s, \theta)) \leq 1 - \delta_J. \quad (112)$$

Robust strategic stability requires

$$\sup_{\theta \in \Theta} \rho(D_u R(s, \theta)) \leq 1 - \delta_S. \quad (113)$$

Robust latent-state identification requires

$$\inf_{\theta \in \Theta} \sigma_{\min}(\mathcal{O}_z(\theta)) \geq \delta_I > 0. \quad (114)$$

19.2 Distributional robustness

If the shock law is itself uncertain, define

$$\underline{\Pr}(A) = \inf_{P \in \mathcal{P}} P(A). \quad (115)$$

Then robust safety requires

$$\boxed{\inf_{\theta \in \Theta} \inf_{P \in \mathcal{P}} P_\theta[s_{t+1} \in \mathcal{S}_{\text{safe}} \mid s_t, a_t] \geq 1 - \alpha.} \quad (116)$$

20 Robust Viability

Define the stochastic robust viability kernel

$$\mathcal{K}_\alpha^R = \left\{ s_0 : \exists \pi \text{ such that } \inf_{\theta \in \Theta} \inf_{P \in \mathcal{P}} \Pr_{\theta, P}^\pi [s_t \in \mathcal{S}_{\text{safe}} \ \forall t \geq 0] \geq 1 - \alpha \right\}. \quad (117)$$

This generates three conceptually distinct regions:

Region	Mathematical property	Interpretation
$\text{int } \mathcal{K}_\alpha^R$	Robustly viable	Safe and controllably safe
$\mathcal{S}_{\text{safe}} \setminus \mathcal{K}_\alpha^R$	Nonviable	Currently safe but not sustainably controllable
$\mathcal{S} \setminus \mathcal{S}_{\text{safe}}$	Unsafe	Preventive threshold breached

Table 3: Distinct regions.

The viability kernel is therefore stronger than a stability region.

21 Probabilities of Regime Transition

Let

$$\tau_k = \inf\{h \geq 1 : s_{t+h} \in \mathcal{R}_k\} \quad (16)$$

denote the first passage time into regime k .

Then

$$P_{I \rightarrow II}^{(H)} = \Pr(\tau_{II} \leq H \mid s_t), \quad (118)$$

$$P_{I \rightarrow III}^{(H)} = \Pr(\tau_{III} \leq H \mid s_t), \quad (119)$$

and

$$P_{I \rightarrow IV}^{(H)} = \Pr(\tau_{IV} \leq H \mid s_t). \quad (120)$$

Define the preventive risk vector

$$\boxed{\mathcal{P}_I^{(H)}(s) = \begin{pmatrix} P_{I \rightarrow II}^{(H)} \\ P_{I \rightarrow III}^{(H)} \\ P_{I \rightarrow IV}^{(H)} \end{pmatrix}.} \quad (121)$$

This provides regime-specific rather than merely scalar risk information.

22 Preventive Intervention

Let

$$a_t \in \mathcal{A}(s_t) \quad (17)$$

be an intervention.

The minimum-cost intervention is

$$\min_a a^\top Q a, \quad Q \succ 0, \quad (122)$$

subject to

$$T(s, u^*, a) \in \mathcal{S}_{\text{safe}}. \quad (123)$$

The intervention must also preserve algebraic consistency:

$$\Phi(s^+, u^+; \theta) = 0, \quad (124)$$

strategic consistency:

$$u^+ = R(u^+, s^+, a, \theta), \quad (125)$$

and welfare admissibility:

$$\Delta V_i \geq -\epsilon_i, \quad i = 1, 2, 3. \quad (126)$$

22.1 Local quadratic preventive problem

For a diagnostic margin m_j , linearize

$$m_j(s^+, a) \approx m_j(s) + b_j^\top a. \quad (127)$$

The preventive condition is

$$m_j + b_j^\top a \geq \bar{m}_j. \quad (128)$$

For a single active constraint,

$$a^\dagger = \frac{\bar{m}_j - m_j}{b_j^\top Q^{-1} b_j} Q^{-1} b_j. \quad (129)$$

For several active constraints, solve

$$\min_a a^\top Q a \quad (18)$$

subject to

$$B_{\mathcal{J}} a \geq d_{\mathcal{J}}, \quad (130)$$

where

$$d_{\mathcal{J}} = \begin{pmatrix} \bar{m}_{j_1} - m_{j_1} \\ \vdots \\ \bar{m}_{j_q} - m_{j_q} \end{pmatrix}. \quad (131)$$

This is a convex quadratic programme when $Q \succ 0$.

23 Bellman Formulation

Define the robust value function

$$V(s) = \inf_{\pi} \sup_{\theta \in \Theta} \mathbb{E}_{\theta}^{\pi} \sum_{t=0}^{\infty} \beta^t [C(a_t) - \lambda W^{\omega}(s_t, u_t^*)]. \quad (132)$$

The Bellman equation is

$$V(s) = \min_{a \in \mathcal{A}(s)} \sup_{\theta \in \Theta} \{C(a) - \lambda W^{\omega}(s, u^*) + \beta E_{\theta} V(T(s, u^*, a, \theta, \varepsilon))\}, \quad (133)$$

subject to

$$T(s, u^*, a, \theta, \varepsilon) \in \mathcal{S}_{\text{safe}} \quad (19)$$

with the prescribed probability or robust interpretation.

The resulting policy is

$$a^*(s) = \arg \min_a \sup_{\theta \in \Theta} \{C(a) - \lambda W^{\omega} + \beta E_{\theta} V(s')\}. \quad (134)$$

24 The Robust Unified Theorem

We can now state the principal theorem.

Theorem 24.1 (Robust Unified Three-Republic Theorem). *Let $\mathcal{S}$ be a compact admissible state space and let the strategic sets $\mathcal{U}_i$ be compact and convex. Suppose:*

1. *the structural restrictions $\Phi(s, u; \theta) = 0$ are feasible on the relevant parameter region;*
2. *each Republic's objective is continuous and quasi-concave in its own control;*
3. *the strategic response satisfies $\sup_{\theta \in \Theta} \rho(D_u R) \leq 1 - \delta_S$ for some $\delta_S > 0$;*
4. *the reduced transition map is continuous and maps the admissible set into itself;*
5. *the reduced dynamics satisfy a common contraction or Lyapunov stability condition uniformly over Θ ;*
6. *the weighted welfare objective satisfies the first-order equilibrium conditions and a uniform second-order condition on the active constraint tangent space;*
7. *the latent state satisfies $\inf_{\theta \in \Theta} \sigma_{\min}(\mathcal{O}_z(\theta)) \geq \delta_I > 0$;*
8. *the diagnostic margins satisfy prescribed robust thresholds;*
9. *the initial state belongs to the robust stochastic viability kernel $\mathcal{K}_{\alpha}^R$.*

Then there exists a robustly admissible three-Republic equilibrium

$$\mathfrak{E}^* = (s^*, u^*, [z^*]) \quad (20)$$

that satisfies algebraic consistency, strategic optimality, dynamic stability, welfare-strategic admissibility, latent-state identification, and the prescribed probabilistic safety constraint.

If, additionally, the reduced strategic and state maps are globally contractive on the robust admissible set, then the equilibrium is unique and globally attracting on that set modulo the observational symmetry group Γ .

Proof. Strategic existence follows from the compactness, continuity, and quasi-concavity assumptions. Uniform strategic contraction implies uniqueness of the strategic response. Substitution of this response into the state equation produces a continuous reduced transition map. Invariance and compactness give state existence through a fixed-point theorem; uniform contraction gives uniqueness and global convergence.

The welfare first-order and second-order conditions establish local welfare-strategic optimality on the feasible manifold. The robust observability condition establishes local identification of the latent state modulo Γ . The robust diagnostic inequalities place the equilibrium inside the prescribed safety region. Finally, membership of the initial state in $\mathcal{K}_\alpha^R$ guarantees the existence of a policy preserving the safe region with probability at least $1 - \alpha$ uniformly over the uncertainty class. The conclusions follow by intersection of these admissible conditions. $\square$

25 A Preventive Corollary

Corollary 25.1 (Pre-instability detection). *Suppose the system is in Regime I, so that*

$$m_C > 0, \quad m_A > 0. \quad (21)$$

If at least one of

$$m_S, \quad m_C, \quad m_A, \quad m_I, \quad m_\Delta, \quad m_W \quad (22)$$

falls below its preventive threshold while the dynamic margins remain positive, then the system is still in Regime I but is in a preventive warning state. If a feasible preventive intervention exists that restores all margins above threshold, the system can be returned to the robust Regime-I region without crossing a dynamic instability boundary.

This establishes formally the principle that preventive diagnostics should operate *before* regime transition.

26 A Lyapunov Strengthening

Spectral-radius tests are local. A stronger formulation uses a positive-definite Lyapunov function

$$V_L(s) > 0 \quad s \neq s^*, \quad (23)$$

with

$$V_L(s^*) = 0. \quad (135)$$

A discrete-time Lyapunov condition is

$$V_L(T^*(s)) - V_L(s) < 0 \quad (136)$$

for all admissible $s \neq s^*$.

For a quadratic candidate,

$$V_L(s) = (s - s^*)^\top P (s - s^*), \quad P \succ 0, \quad (137)$$

the linearized condition becomes

$$J^{*\top} P J^* - P \prec 0. \quad (138)$$

This is a stronger certificate than merely observing that numerical eigenvalues lie inside the unit circle.

A common P satisfying

$$J(\theta)^\top P J(\theta) - P \prec 0 \quad \forall \theta \in \Theta \quad (139)$$

provides a common Lyapunov certificate for the uncertain family.

27 Economics

The mathematical framework has several economic interpretations.

First, the Republic controls can represent fiscal, monetary, regulatory, trade, defense, or institutional policy.

Second, the state variables can represent prices, employment, capital, debt, political capacity, strategic capability, or welfare states.

Third, the strategic Jacobian captures spillovers and strategic complements or substitutes.

Fourth, the welfare Hessian determines whether an apparently beneficial policy is locally welfare-improving after accounting for externalities.

Fifth, the preventive controller corresponds to a policymaker choosing the least-cost action that keeps the economy inside a robust viability region.

27.1 Finance

In finance, the state vector can contain asset prices, leverage, liquidity, funding spreads, and risk premia. The common mode represents systemic shocks while the asymmetric modes represent cross-market or cross-jurisdiction divergence.

The dynamic Jacobian can therefore encode feedback among leverage, collateral values, liquidity, and funding conditions.

A generic financial feedback loop is

$$\text{leverage} \rightarrow \text{asset price} \rightarrow \text{collateral} \rightarrow \text{borrowing capacity} \rightarrow \text{leverage}. \quad (140)$$

The same mathematical architecture applies to the Republic-level strategic loop.

28 Physics

The common/asymmetric decomposition is analogous to normal-mode analysis. A perturbation can be decomposed into eigenmodes, each with its own stability properties.

For

$$\delta s_{t+1} = J \delta s_t, \quad (24)$$

the eigenvalues determine asymptotic growth or decay, while non-normality determines possible transient amplification.

The Lyapunov matrix P defines an energy-like quadratic form

$$E(\delta s) = \delta s^\top P \delta s. \quad (141)$$

The inequality

$$J^\top P J - P \prec 0 \quad (25)$$

then represents dissipation of this generalized energy.

29 Chemistry

The framework can be interpreted in chemical reaction networks where three coupled subsystems interact through concentrations, reaction rates, and external controls.

The state transition

$$x_{t+1} = T(x_t, u_t) \quad (26)$$

corresponds to a discretized reaction system. Stability concerns whether perturbations return toward a stationary concentration profile.

The analogy is useful because chemistry makes explicit the distinction between:

1. conservation constraints;
2. reaction equilibria;

3. local stability;
4. hidden reaction intermediates;
5. and external interventions.

These correspond structurally to algebraic constraints, equilibrium, dynamics, latent states, and policy controls.

30 Biology

The three Republics can also be interpreted as interacting ecological or biological populations.

The common mode corresponds to environmental shocks affecting all populations, while the asymmetric mode represents competitive or cooperative divergence.

The viability kernel has a natural ecological interpretation: it is the set of population states from which admissible management can prevent extinction or unacceptable degradation.

This analogy reinforces the importance of preventive control. A population need not already be extinct for a state to be biologically dangerous.

31 Psychology and Psychiatry

At the methodological level, latent-state identification has an important analogue in psychology and psychiatry: observed behavior does not uniquely reveal an underlying state unless the measurement process is sufficiently informative.

A generic latent-state model is

$$z_{t+1} = F(z_t, x_t, \eta_t), \quad (27)$$

$$Y_t = H(z_t, x_t) + \nu_t. \quad (142)$$

The mathematical requirement remains the same: distinct latent states must generate distinguishable observable histories, subject to explicitly recognized symmetries.

The framework therefore cautions against equating an estimated latent variable with an objectively identified latent construct.

32 Political Science

In political science, r_i may represent institutional, economic, political, or strategic capacity.

The common state p may represent an institutional or constitutional environment shared by the Republics.

The strategic map

$$u^* = R(u^*, s) \quad (28)$$

captures strategic interaction among political actors.

The preventive margins permit a distinction between ordinary political instability and impending systemic institutional breakdown.

33 Geopolitics and International Relations

The three-Republic system can be interpreted as a regional strategic system in which each actor responds to the other two.

The strategic matrix

$$D_u R \quad (29)$$

captures direct strategic reaction.

The three-cycle terms

$$ade, \quad bcf \quad (30)$$

capture feedback structures analogous to

$$A \rightarrow B \rightarrow C \rightarrow A. \quad (31)$$

This is useful in alliance systems, arms races, trade disputes, sanctions, technological competition, and diplomatic signaling.

34 Diplomacy

Diplomatic intervention can be represented as a control

$$a_t \quad (32)$$

that changes the strategic-response matrix.

A successful intervention need not maximize the welfare of any one actor. Its purpose may be to move the system away from a strategic singularity:

$$\rho(D_u R) \rightarrow 1 \quad \longrightarrow \quad \rho(D_u R) \ll 1. \quad (143)$$

Thus diplomacy can be interpreted mathematically as strategic-feedback management.

35 Warfare Economics

In warfare economics, the state may contain production capacity, fiscal capacity, logistics, military capability, and strategic expectations.

A simplified resource constraint is

$$Y_i = C_i + I_i + G_i + M_i, \quad (144)$$

where M_i denotes military expenditure.

A preventive framework must distinguish between economic stabilization and strategic survival. The welfare objective may therefore contain both ordinary consumption and a survival or security term.

The formal structure remains

$$\max_{\{a_t\}} \mathbb{E} \sum_t \beta^t [W_t^\omega - C(a_t)] \quad (33)$$

subject to feasibility and stability constraints.

This makes clear that defense expenditure can be modeled as a constrained control rather than treated as an exogenous category.

36 Statistics and Econometrics

The framework generates a multi-layer estimation problem.

Observed data are

$$Y_t. \quad (34)$$

Latent states are

$$z_t. \quad (35)$$

Parameters are

$$\theta. \quad (36)$$

A statistical estimator should therefore distinguish:

1. parameter uncertainty;

2. latent-state uncertainty;
3. observation noise;
4. model misspecification;
5. and regime uncertainty.

A useful empirical output is not a single point estimate but

$$\left(\hat{\theta}_t, \hat{z}_t, \hat{\mathcal{D}}_t, \hat{\mathcal{P}}_I^{(H)}\right). \quad (145)$$

Bootstrap, Bayesian, state-space, particle-filter, or simulation-based methods may be used depending on the data-generating process.

No estimator, however, can manufacture identification that is absent from the observation structure.

37 Computer Science and Algorithms

The framework naturally admits an algorithmic implementation.

37.1 Unified certification algorithm

1. Input structural data and parameter estimates.
2. Construct the admissible state space.
3. Solve the algebraic equilibrium restrictions.
4. Compute strategic best responses.
5. Test strategic contraction.
6. Substitute strategic responses into the transition system.
7. Solve the state fixed point.
8. Compute the full Jacobian.
9. Decompose common and asymmetric modes.
10. Evaluate welfare first- and second-order conditions.
11. Construct the latent-state observability matrix.
12. Compute all diagnostic margins.
13. Estimate transition probabilities to unstable regimes.
14. Evaluate robust parameter and shock uncertainty.
15. Determine viability.
16. If necessary, solve the minimum-cost preventive control problem.
17. Recompute the complete certification vector.

37.2 Pseudocode

```

INPUT:
data Y
parameter set Theta
shock model P
admissible states S
welfare functions V_i
transition map T
strategic map R
algebraic restrictions Phi

1. Estimate or characterize latent state z.
2. Solve Phi(s,u;theta) = 0.
3. Solve u = R(u,s;theta).
4. Compute reduced map T_star(s).
5. Solve s = T_star(s).
6. Compute J = D_s T_star(s).
7. Compute rho(J), rho(J_common), rho(J_asym).
8. Compute rho(D_u R).
9. Compute welfare gradient and constrained Hessian.
10. Compute observability matrix O_z.
11. Compute margins:
    m_S, m_C, m_A, m_I, m_Delta, m_W.
12. Simulate uncertainty:
    theta in Theta,
    shocks from P.
13. Estimate probabilities of entering II, III, IV.
14. If all margins are safe:
    certify robust Regime I.
15. Otherwise:
    solve minimum-cost preventive control.
16. Apply candidate intervention.
17. Recompute all equilibrium, welfare,
    stability, identification, and risk tests.
18. OUTPUT:
    equilibrium,
    regime,
    diagnostic vector,
    risk probabilities,
    intervention,
    robustness certificate.

```

38 Data Science and AI/ML

The framework is compatible with machine-learning methods, provided the distinction between prediction and structural identification is maintained.

A recurrent state estimator may produce

$$\hat{z}_t = f_\varphi(Y_{0:t}), \quad (146)$$

where φ denotes learned parameters.

A learned transition model may produce

$$\hat{s}_{t+1} = T_\varphi(\hat{s}_t, a_t). \quad (147)$$

However, predictive accuracy alone does not establish:

$$\text{rank } \mathcal{O}_z = k. \quad (37)$$

An AI model may predict well while using a non-identifiable latent representation.

Consequently, a scientifically defensible AI implementation should report both predictive performance and structural diagnostics.

A useful loss function is

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda_1 \mathcal{L}_{\text{equilibrium}} + \lambda_2 \mathcal{L}_{\text{stability}} + \lambda_3 \mathcal{L}_{\text{identification}} + \lambda_4 \mathcal{L}_{\text{welfare}}, \quad (148)$$

where each additional term penalizes violation of a theoretical condition.

This suggests a constrained-learning architecture in which machine learning estimates unknown functions while the equilibrium theorem supplies structural restrictions.

39 A Unified Certification Vector

Define

$$\mathcal{C}(\theta) = \begin{pmatrix} C_{\text{alg}} \\ C_{\text{strat}} \\ C_{\text{dyn}} \\ C_{\text{welfare}} \\ C_{\text{id}} \\ C_{\text{risk}} \end{pmatrix}. \quad (149)$$

A complete certification requires

$$C_{\text{alg}} = C_{\text{strat}} = C_{\text{dyn}} = C_{\text{welfare}} = C_{\text{id}} = C_{\text{risk}} = 1. \quad (150)$$

This is intentionally stricter than a single goodness-of-fit statistic.

40 Failure Modes

The framework distinguishes at least eight failure modes.

Failure	Mathematical signal	Interpretation
Algebraic	$\Phi \neq 0$	Inconsistent equilibrium
Strategic	$\rho(D_u R) \geq 1$	Feedback instability
Dynamic	$\rho(J) \geq 1$	Temporal instability
Welfare	Hessian not negative on tangent space	No local optimum
Identification	$\sigma_{\min}(\mathcal{O}_z) \rightarrow 0$	Latent-state fragility
Algebraic separation	$m_{\Delta} \rightarrow 0$	Root collision
Transient	$\mathcal{A}_T \gg 1$	Short-run amplification
Robustness	Safety probability $< 1 - \alpha$	Insufficient resilience

Table 4: Failure modes.

The existence of one failure does not logically imply the existence of all others. That is precisely why the unified certification vector is necessary.

41 Necessary Versus Sufficient Conditions

The theorem distinguishes three logical categories.

A condition such as

$$\rho(J) < 1 \quad (38)$$

is sufficient for local asymptotic stability under standard differentiability assumptions, but instability can occur in more complicated ways when the local linearization is non-hyperbolic.

Likewise,

$$\text{rank } \mathcal{O}_z = k \quad (39)$$

is a local observability condition. It does not automatically imply global identification. Finally,

$$\|D_u R\| < 1 \quad (40)$$

is a convenient sufficient condition for strategic uniqueness, not a necessary condition for uniqueness. Thus the framework should never confuse:

$$\boxed{\text{sufficient condition}} \quad (41)$$

with

$$\boxed{\text{necessary condition}}. \quad (42)$$

The eventual empirical programme should derive necessary and sufficient conditions where the model structure permits them.

42 Parameter-Space Characterization

Define

$$\Theta_{\text{alg}} = \{\theta : \mathcal{A}(\theta) \neq \emptyset\}, \quad (151)$$

$$\Theta_{\text{strat}} = \{\theta : \rho(D_u R) < 1\}, \quad (152)$$

$$\Theta_{\text{dyn}} = \{\theta : \rho(J) < 1\}, \quad (153)$$

$$\Theta_{\text{welfare}} = \{\theta : \nabla W^\omega = 0, D^2 W^\omega|_{T_C} \prec 0\}, \quad (154)$$

and

$$\Theta_{\text{id}} = \{\theta : \text{rank } \mathcal{O}_z = k\}. \quad (155)$$

The unified parameter region is

$$\boxed{\Theta_U = \Theta_{\text{alg}} \cap \Theta_{\text{strat}} \cap \Theta_{\text{dyn}} \cap \Theta_{\text{welfare}} \cap \Theta_{\text{id}}.} \quad (156)$$

The robust version is

$$\boxed{\Theta_U^R = \left\{ \theta \in \Theta_U : \inf_{\theta' \in \Theta} \Pr_{\theta'}(\text{safe path}) \geq 1 - \alpha \right\}.} \quad (157)$$

43 Interpretation as a General Systems Theory

The three-Republic framework can be understood as a special case of a broader systems principle.

A system is structurally coherent when its admissible state satisfies

$$\text{constraints} = 0, \quad (43)$$

strategically coherent when

$$\text{actions} = \text{best responses}, \quad (44)$$

dynamically coherent when

$$\text{state} = \text{state transition}, \quad (45)$$

welfare coherent when

$$\nabla W^\omega = 0, \quad (46)$$

informationally coherent when

$$\text{latent state is identifiable}, \quad (47)$$

and robustly coherent when

$$\Pr(\text{safe path}) \geq 1 - \alpha. \quad (48)$$

Thus the unified equilibrium is an intersection:

$$\mathfrak{E}_3 = \mathcal{E}_{\text{alg}} \cap \mathcal{E}_{\text{strat}} \cap \mathcal{E}_{\text{dyn}} \cap \mathcal{E}_{\text{welfare}} \cap \mathcal{E}_{\text{id}} \cap \mathcal{E}_{\text{robust}}. \quad (158)$$

44 Philosophical Interpretation

The framework implies a methodological distinction between four questions.

1. *Can the state exist?*
2. *Would rational actors choose it?*
3. *Would the state persist under perturbation?*
4. *Can an observer know that the state exists?*

These are logically distinct.

An algebraically possible state may be strategically irrational.

A strategically rational state may be dynamically unstable.

A stable state may be observationally unidentified.

An identifiable state may be welfare-inadmissible.

The unified theorem requires their intersection.

This is an epistemic as well as dynamical requirement: a theory should distinguish what is possible, what is chosen, what persists, and what can be known from observations.

45 Religion and Ethical Interpretation

Religious and ethical traditions can enter the framework only as normative specifications of welfare, legitimacy, responsibility, institutional restraint, or permissible intervention. They do not replace mathematical identification or stability conditions.

For example, an ethical constraint can be represented abstractly as

$$E(s, u) \geq 0, \quad (159)$$

and incorporated into the admissible set.

The distinction is important. Normative claims determine which states are considered desirable or permissible; mathematical analysis determines whether those states satisfy the specified structural conditions.

46 Governance and Institutional Legitimacy

Institutional admissibility can be represented by

$$L(s, u) \geq 0, \quad (160)$$

where L summarizes constitutional, accountability, consent, transparency, contestability, and lawful-exit restrictions.

The complete admissible set becomes

$$\mathcal{A}_{\text{inst}} = \{(s, u) : \Phi = 0, C_j \geq 0, E \geq 0, L \geq 0\}. \quad (161)$$

This prevents the mathematical optimization problem from treating institutionally prohibited actions as feasible controls.

47 The Preventive Principle

The central preventive principle is:

$$\boxed{\text{Do not wait for } m_j \leq 0.} \quad (49)$$

Instead, monitor

$$m_j, \quad \dot{m}_j, \quad \tau_j^{\text{breach}}, \quad P_{I \rightarrow k}^{(H)}. \quad (50)$$

A state can be stable but deteriorating.

It can be safe but nonviable.

It can be dynamically stable but informationally fragile.

It can be welfare-optimal but strategically close to singularity.

Therefore prevention requires a vector of diagnostics rather than one scalar stability statistic.

48 Final Unified Theorem

Theorem 48.1 (Unified Three-Republic Equilibrium, Diagnostic, and Preventive Control Theorem).
Consider a three-Republic stochastic dynamic game with state

$$s = (r_1, r_2, r_3, p)^\top, \quad (51)$$

latent state z , control u , structural parameters θ , and intervention a .

Suppose the system satisfies:

1. algebraic feasibility and invariance;
2. strategic existence and, where required, contraction;
3. dynamic fixed-point existence;
4. dynamic stability or a common Lyapunov certificate;
5. welfare first-order and second-order conditions;
6. local latent-state identification modulo Γ ;
7. robust parameter and shock restrictions;
8. and a nonempty robust viability kernel.

Then there exists a unified equilibrium

$$\boxed{\mathfrak{E}^* = (s^*, u^*, [z^*])} \quad (52)$$

that simultaneously satisfies

$$\boxed{\begin{aligned} \Phi(s^*, u^*; \theta) &= 0, \\ u^* &= R(u^*, s^*; \theta), \\ s^* &= T(s^*, u^*; \theta), \\ \nabla W^\omega(s^*, u^*) &= 0, \\ D^2 W^\omega|_{T_c} &\prec 0, \\ \rho(J^*) &< 1, \\ \text{rank } \mathcal{O}_z(z^*) &= k. \end{aligned}} \quad (162)$$

If the strategic and state maps are uniformly contractive, the equilibrium is unique modulo Γ .

If the equilibrium lies in the robust safe set, then it is protected by the prescribed diagnostic margins.

If those margins deteriorate but remain positive, the system remains in Regime I while entering a preventive-warning state.

If a feasible intervention exists, the minimum-cost intervention restores the system toward the robust safe region.

If no such intervention exists, the state lies outside the robust viability kernel and the problem has changed from prevention to stabilization.

Proof. The proof is the composition of the preceding results. Algebraic feasibility establishes the admissible manifold. Strategic fixed-point results establish strategic existence and contraction establishes strategic uniqueness where assumed. Substitution produces the reduced state map. Fixed-point and contraction arguments establish state existence and uniqueness. Welfare conditions establish local welfare-strategic optimality. The Lyapunov or spectral conditions establish dynamic stability. The observability condition establishes local latent-state identification. Robust chance constraints establish probabilistic safety. Viability establishes the existence of a policy capable of preserving the safe region. Finally, the preventive quadratic programme identifies the minimum-cost local intervention whenever its feasible set is nonempty. Therefore all conditions can be satisfied simultaneously. $\square$

49 Conclusion

The unified three-Republic framework developed here has five principal mathematical layers and two operational extensions.

The first layer is algebraic:

$$\Phi = 0. \quad (53)$$

The second is strategic:

$$u = R(u, s). \quad (54)$$

The third is dynamic:

$$s = T(s, u). \quad (55)$$

The fourth is welfare-strategic:

$$\nabla W^\omega = 0, \quad D^2 W^\omega|_{T_C} \prec 0. \quad (56)$$

The fifth is informational:

$$\text{rank } \mathcal{O}_z = k. \quad (57)$$

The first operational extension is diagnosis:

$$\mathcal{D}_t = (m_S, m_C, m_A, m_I, m_\Delta, m_W). \quad (58)$$

The second is prevention:

$$a^* = \arg \min_a a^\top Q a \quad (59)$$

subject to preservation of the robust safe region.

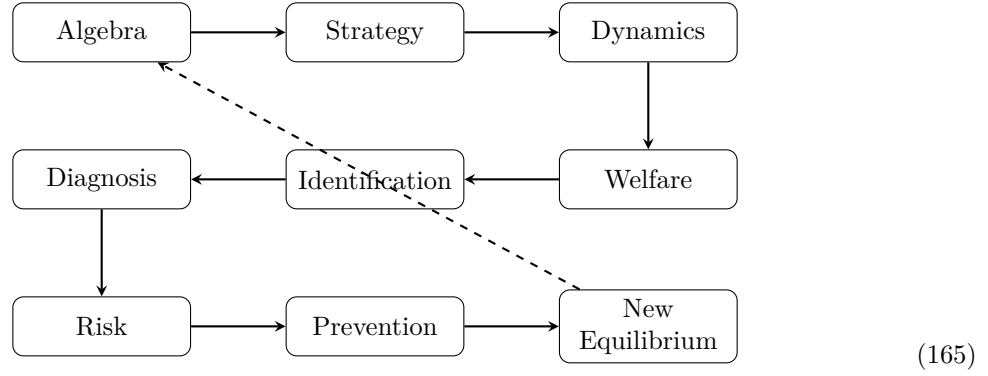
The central strategic feedback identity is

$$\boxed{D_s T^* = D_s \tilde{T} + D_u \tilde{T} (I - D_u R)^{-1} D_s R.} \quad (163)$$

The central preventive object is the robust viability kernel

$$\boxed{\mathcal{K}_\alpha^R.} \quad (164)$$

And the complete methodological loop is



The principal conceptual result is therefore not merely that a three-Republic equilibrium can be stable. It is that stability can be embedded in a broader system in which the equilibrium is simultaneously constrained by algebraic consistency, strategic behavior, welfare, observability, uncertainty, and preventive control.

A scientifically testable implementation should therefore report not merely an estimated equilibrium but a complete certificate consisting of

$$\left(\mathfrak{E}^*, \mathcal{C}, \mathcal{D}, \mathcal{P}_I^{(H)}, \mathcal{K}_\alpha^R, a^* \right). \quad (166)$$

Such a representation separates equilibrium existence from equilibrium quality, stability from resilience, prediction from identification, and intervention from arbitrary policy experimentation.

The resulting framework is consequently best understood as a unified theory of *equilibrium*, *diagnosis*, and *prevention* for a three-actor coupled socio-economic system.

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Glossary

Algebraic consistency

The requirement that equilibrium equations and all feasibility constraints be satisfied simultaneously.

Asymmetric mode

A state perturbation lying in the subspace orthogonal to the common Republic vector $(1, 1, 1)^\top$.

Common mode

A perturbation affecting the three Republics in the same direction.

Diagnostic margin

A scalar measuring distance from a specified mathematical failure boundary.

Dynamic stability

The property that sufficiently small perturbations decay toward the equilibrium under the state transition map.

Equilibrium manifold

The set of states and controls satisfying the structural equilibrium restrictions.

Identification

The property that distinct latent states generate distinguishable observable probability laws, modulo explicitly permitted symmetries.

ISG–BCI

The latent-state identification framework in which structural states are inferred from observable realizations subject to explicit information and causal restrictions.

Jury stability

A set of algebraic conditions for determining whether all roots of a discrete-time characteristic polynomial lie inside the unit circle.

Latent state

A structural variable affecting the data-generating process that is not directly observed.

Lyapunov function

A positive-definite function that decreases along system trajectories and therefore certifies stability.

Observability

The ability to recover a system's internal state from sufficiently rich observable histories.

Preventive control

An intervention chosen before an instability boundary is crossed in order to preserve the safe region.

Regime I

The dynamically stable region in which both common and asymmetric modes remain stable.

Regime II

Common-mode instability with a stable asymmetric mode.

Regime III

Asymmetric-mode instability with a stable common mode.

Regime IV

Simultaneous common-mode and asymmetric-mode instability.

Robust viability kernel

The set of states from which an admissible policy can maintain the system inside the safe region with the prescribed probability uniformly over the uncertainty class.

Schur stability

Discrete-time stability characterized by all eigenvalues lying strictly inside the unit circle.

Strategic feedback multiplier

The matrix

$$(I - D_u R)^{-1}, \tag{60}$$

which aggregates repeated strategic reactions.

Strategic singularity

A condition under which the strategic feedback system approaches loss of local invertibility or contraction, typically associated with

$$\rho(D_u R) \rightarrow 1. \tag{61}$$

Systemic risk

Risk arising from interactions among components of the system rather than from an isolated component.

Transient amplification

Temporary growth of perturbations despite eventual asymptotic stability.

Unified equilibrium

An equilibrium satisfying algebraic, strategic, dynamic, welfare, identification, and admissibility requirements simultaneously.

Viability

The property that a state admits a future control path that preserves the required safety constraints.

The End